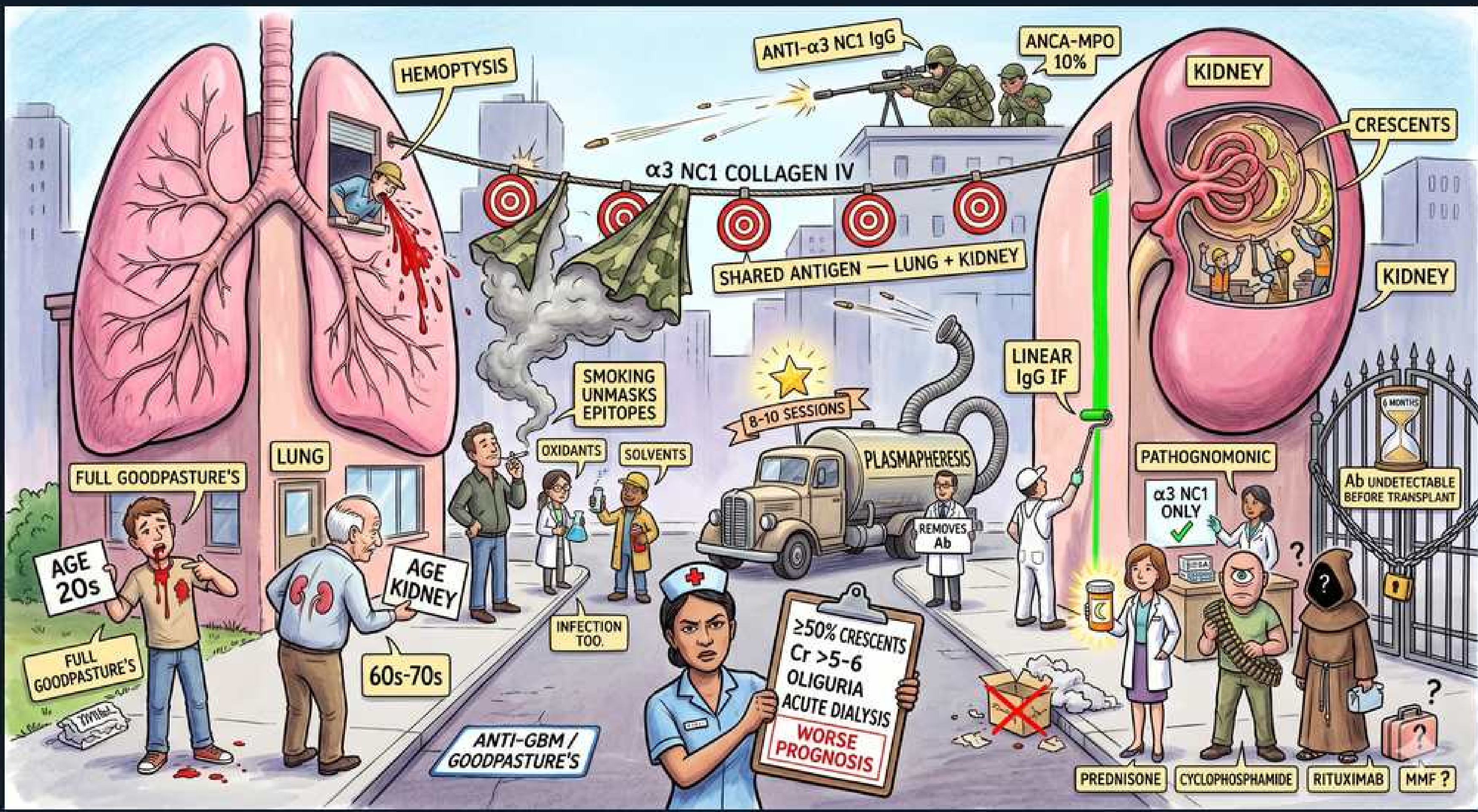
Anti-GBM Disease / Goodpasture Syndrome



1. α3 chain of collagen IV (NC1)

WHAT YOU SEE
Banner 'α3 NC1 collagen IV' across the clothesline

WHAT IT ENCODES
Anti-GBM disease is driven by autoantibodies against the NC1 domain of the α3 chain of type IV collagen (the Goodpasture antigen) in glomerular and alveolar basement membranes. A type II hypersensitivity reaction.

BOARD TRAP
The target is α3(IV)NC1 specifically — Alport's also involves α3/α5 collagen IV mutations, and transplanting an Alport patient can trigger de novo anti-GBM disease.

2. Shared antigen — lung + kidney

WHAT YOU SEE
'Shared antigen — lung + kidney' label, center

WHAT IT ENCODES
The α3(IV) antigen is expressed in both glomerular AND alveolar basement membranes, explaining the pulmonary-renal syndrome: diffuse alveolar hemorrhage plus rapidly progressive GN.

BOARD TRAP
Isolated renal anti-GBM (no lung) is 'anti-GBM disease'; lung + kidney is 'Goodpasture SYNDROME' — but the antibody is identical.

3. Hemoptysis / alveolar hemorrhage

WHAT YOU SEE
Hemoptysis at the window, upper left

WHAT IT ENCODES
Diffuse alveolar hemorrhage causes hemoptysis, falling hemoglobin, and a RAISED DLCO (carbon monoxide bound by intra-alveolar blood). Smokers and those with lung injury are at higher risk for pulmonary involvement.

BOARD TRAP
An ELEVATED DLCO during alveolar hemorrhage is the counterintuitive board pearl — most causes lower DLCO.

4. Smoking unmasks epitopes

WHAT YOU SEE
Smoke / oxidants / solvents cloud, center-left

WHAT IT ENCODES
Lung involvement requires a 'second hit' that exposes alveolar BM epitopes — cigarette smoke, hydrocarbon solvents, oxidants, or infection. Without injury, antibodies may not access alveolar antigen, giving renal-limited disease.

BOARD TRAP
A young smoker with hemoptysis and AKI is the classic Goodpasture vignette — smoking is the lung trigger, not just an incidental habit.

5. Linear IgG on IF

WHAT YOU SEE
'Linear IgG IF' bright green line on kidney, right

WHAT IT ENCODES
Immunofluorescence shows SMOOTH LINEAR ribbon-like IgG (± C3) along the GBM — the hallmark, reflecting antibody binding to a continuous antigen. EM shows no immune deposits.

BOARD TRAP
LINEAR IF = anti-GBM (and diabetic nephropathy nonspecifically); GRANULAR IF = immune-complex GN. Linear vs granular is the single best IF discriminator.

6. Crescentic GN (>50% crescents)

WHAT YOU SEE
Kidney with crescents, right

WHAT IT ENCODES
Light microscopy: diffuse crescentic, necrotizing GN. Anti-GBM characteristically produces extensive crescents (often >50%) of similar age, correlating with rapidly progressive renal failure.

BOARD TRAP
Crescents are the common pathway of all three RPGN types (anti-GBM, immune-complex, pauci-immune ANCA) — IF pattern, not the crescents, tells them apart.

7. >50% crescents = worse prognosis

WHAT YOU SEE
Nurse sign: >50% crescents, Cr>5–6, oliguria, dialysis

WHAT IT ENCODES
Poor prognostic markers: serum creatinine >5–6 mg/dL, oliguria, dialysis-dependence at presentation, and >50–75% crescents. These patients rarely recover renal function even with treatment.

BOARD TRAP
Dialysis-dependence with extensive crescents predicts non-recovery — but still treat the LUNGS, since alveolar hemorrhage responds and is life-threatening.

8. Plasmapheresis (urgent)

WHAT YOU SEE
'Plasmapheresis — removes Ab, 8–10 sessions' truck, center

WHAT IT ENCODES
Treatment triad: urgent PLASMAPHERESIS to remove circulating anti-GBM antibody, plus corticosteroids and cyclophosphamide to halt new antibody production. Daily/alternate-day exchange for ~2 weeks until antibody clears.

BOARD TRAP
Plasmapheresis is MANDATORY and urgent in anti-GBM (unlike its selective role in ANCA) — delaying it costs nephrons and risks fatal hemorrhage.

9. Steroids + cyclophosphamide

WHAT YOU SEE

Prednisone / cyclophosphamide / rituximab / MMF shelf, lower right

WHAT IT ENCODES

Immunosuppression with high-dose corticosteroids + cyclophosphamide suppresses antibody synthesis; rituximab is an alternative. Combined with plasmapheresis this is the standard induction regimen.

BOARD TRAP

Plasmapheresis ALONE is insufficient — antibody rebounds without concurrent immunosuppression to stop production.

---

10. Anti-α3 NC1 IgG only / ANCA overlap

WHAT YOU SEE

'Anti-α3 NC1 IgG, ANCA-MPO 10%' targets, top

WHAT IT ENCODES

Diagnosis: serum anti-GBM antibody (anti-α3(IV)NC1 IgG) by ELISA. About 10–30% are double-positive for ANCA (usually MPO/p-ANCA); these patients behave more like ANCA vasculitis with relapses.

BOARD TRAP

Double-positive (anti-GBM + ANCA) patients need ongoing immunosuppression like ANCA vasculitis — pure anti-GBM is typically monophasic and doesn't relapse.

---

11. Pathognomonic / non-relapsing

WHAT YOU SEE

'Pathognomonic, anti-α3 NC1 ONLY' label, right

WHAT IT ENCODES

Pure anti-GBM disease is typically a MONOPHASIC, non-relapsing illness — once antibody is cleared and immunosuppression completed, recurrence is rare (unlike ANCA vasculitis). Antibody titer guides duration of therapy.

BOARD TRAP

Don't commit pure anti-GBM patients to indefinite maintenance immunosuppression — it's monophasic; long maintenance is for the ANCA-overlap subset.

---

12. Antibody undetectable before transplant

WHAT YOU SEE

'Ab undetectable before transplant' note, far right

WHAT IT ENCODES

Transplantation should be deferred until anti-GBM antibodies are undetectable for ~6–12 months to avoid recurrent disease in the allograft. Confirm seronegativity before listing.

BOARD TRAP

Transplanting while anti-GBM titers are still positive risks aggressive recurrence in the graft — antibody clearance is a prerequisite.
